

$$v_0 = 100 \frac{m}{s}$$

$$m = 97000 \text{ kg}$$

$$F = -5v^2 - 570000$$

$$m \frac{dv}{dt} = -5v^2 - 570000$$

$$97000 \frac{dv}{dt} = -5v^2 - 570000$$

a)

$$m \frac{dv}{dt} = -5v^2 - 570000 \Leftrightarrow \frac{dv}{dt} = \frac{-5v^2 - 570000}{m}$$

$$dv = \frac{-5v^2 - 570000}{m} dt = \frac{m}{5v^2 + 570000} dt$$

$$t_E = \int_0^{100} \frac{97000}{5v^2 + 570000} dv$$

$$t_E \approx 16,545 \text{ s}$$

b) $\frac{dx}{dt} = v \quad dx = v dt$

$$dx = v \cdot \left(-\frac{m}{5v^2 + 570000} \right) dv \Leftrightarrow dx = -\frac{mv}{5v^2 + 570000} dv$$

$$x_E = \int_{400}^0 -\frac{97000 v}{5v^2 + 570000} dv$$

$$x_E = \int_0^{100} \frac{97000 v}{5v^2 + 570000} dv$$

$$x_E \approx 825,606 \text{ m}$$

Python code

```
import numpy as np

def trapezoidal_rule(function_to_integrate, lower_bound, upper_bound, number_of_intervals):
    interval_width = (upper_bound - lower_bound) / number_of_intervals

    total_sum = 0.5 * function_to_integrate(lower_bound)
    total_sum += 0.5 * function_to_integrate(upper_bound)

    for interval_index in range(1, number_of_intervals):
        current_value = lower_bound + interval_index * interval_width
        total_sum += function_to_integrate(current_value)

    return interval_width * total_sum

def romberg_extrapolation(function_to_integrate, lower_bound, upper_bound, romberg_levels):
    romberg_table = np.zeros((romberg_levels, romberg_levels))

    for row_index in range(romberg_levels):
        number_of_intervals = 2 ** row_index
        romberg_table[row_index, 0] = trapezoidal_rule(
            function_to_integrate,
            lower_bound,
            upper_bound,
            number_of_intervals
        )

    for column_index in range(1, romberg_levels):
        for row_index in range(column_index, romberg_levels):
            factor = 4 ** column_index
            romberg_table[row_index, column_index] = (
                factor * romberg_table[row_index, column_index - 1]
                - romberg_table[row_index - 1, column_index - 1]
            ) / (factor - 1)

    return romberg_table

airplane_mass = 97000
initial_velocity = 100
romberg_levels = 6

def time_integrand(velocity):
    return airplane_mass / (5 * velocity ** 2 + 570000)

def distance_integrand(velocity):
    return airplane_mass * velocity / (5 * velocity ** 2 + 570000)

time_romberg_table = romberg_extrapolation(
    time_integrand,
    0,
    initial_velocity,
    romberg_levels
)

distance_romberg_table = romberg_extrapolation(
    distance_integrand,
    0,
    initial_velocity,
    romberg_levels
)

stopping_time = time_romberg_table[-1, -1]
braking_distance = distance_romberg_table[-1, -1]

print("Romberg Tabelle t_E:")
print(time_romberg_table)
print()

print("Romberg-Tabell x_E:")
print(distance_romberg_table)
print()

print(f"Bremszeit t_E = {stopping_time:.6f} s")
print(f"Bremsweg x_E = {braking_distance:.6f} m")
```